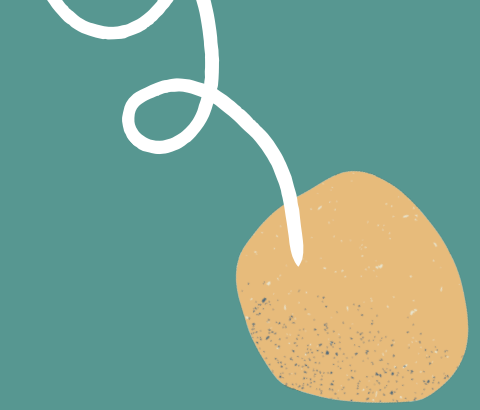


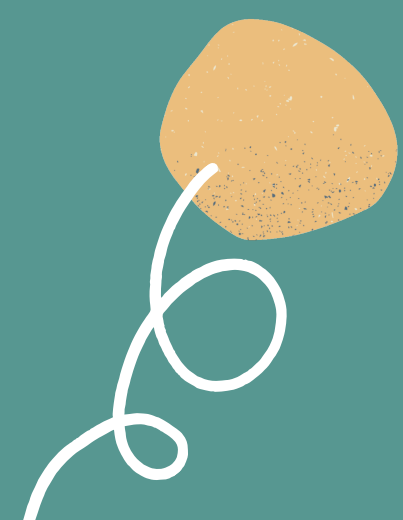
The Compliance Bottleneck for Indian MSEs

- *The Struggle:* Indian Micro and Small Enterprises (MSEs) lose weeks navigating complex, jargon-heavy Bureau of Indian Standards (BIS) documents.
- *The Layout Trap:* Standards are buried in dense, multi-column PDFs that make manual search tedious and error-prone.
- *The Cost:* Misidentifying a standard leads to rejected products, wasted manufacturing cycles, and delayed market entry.
- *The Goal:* Accelerate compliance by turning simple product descriptions into precise BIS standard recommendations instantly.





BIS Copilot: Instant & Accurate Standard Discovery



01

Natural Language Input: Users simply type what they manufacture.

02

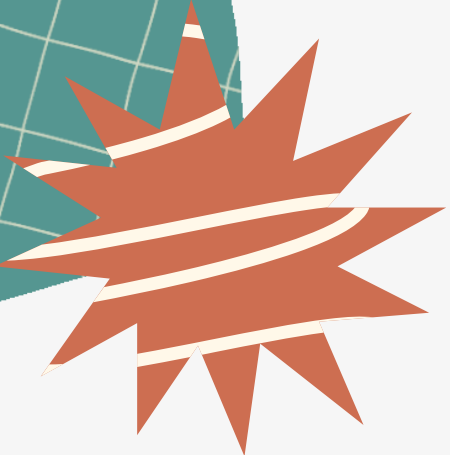
Lightning Fast: Returns the top 5 applicable standards in under 100 milliseconds.

03

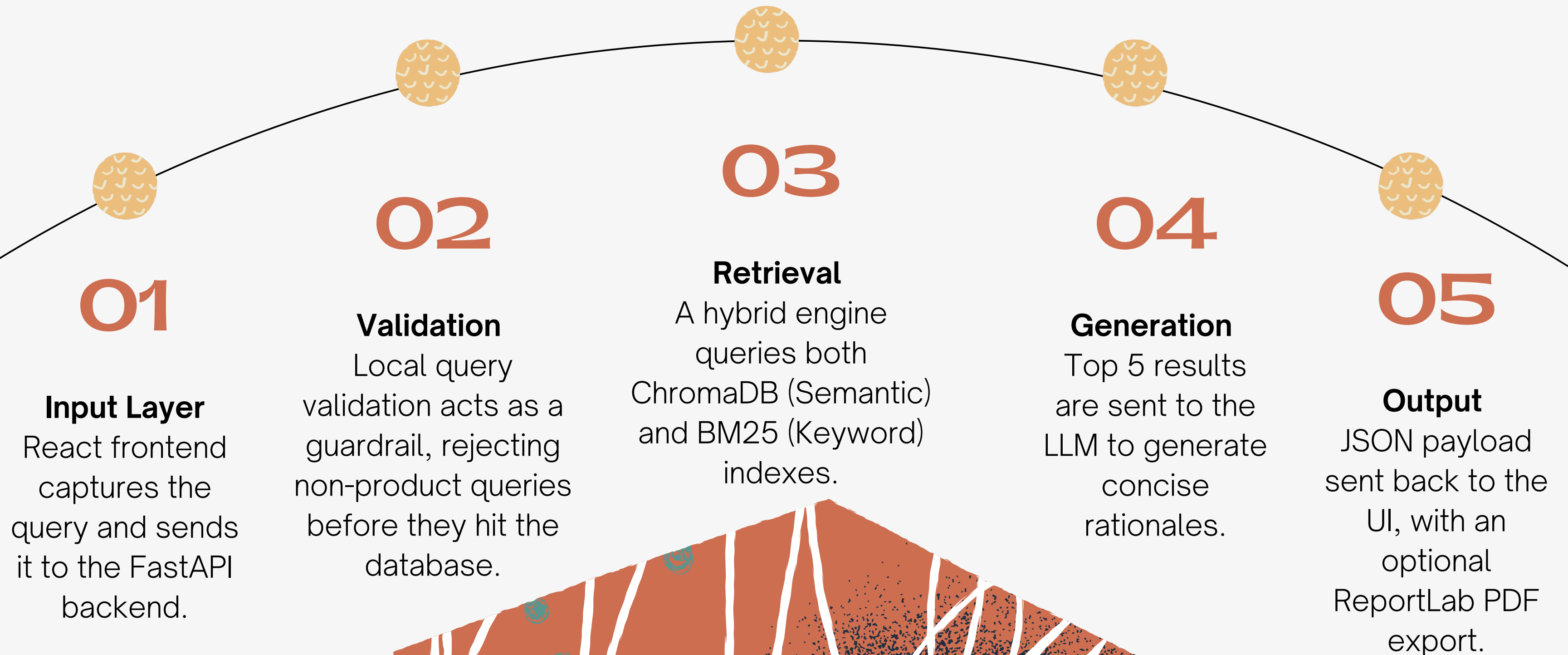
AI Rationale: Doesn't just give a code; it provides a plain-English, one-sentence explanation of why the standard applies.

04

Actionable Deliverable: Generates a downloadable, professional PDF compliance report for the business owner with one click.



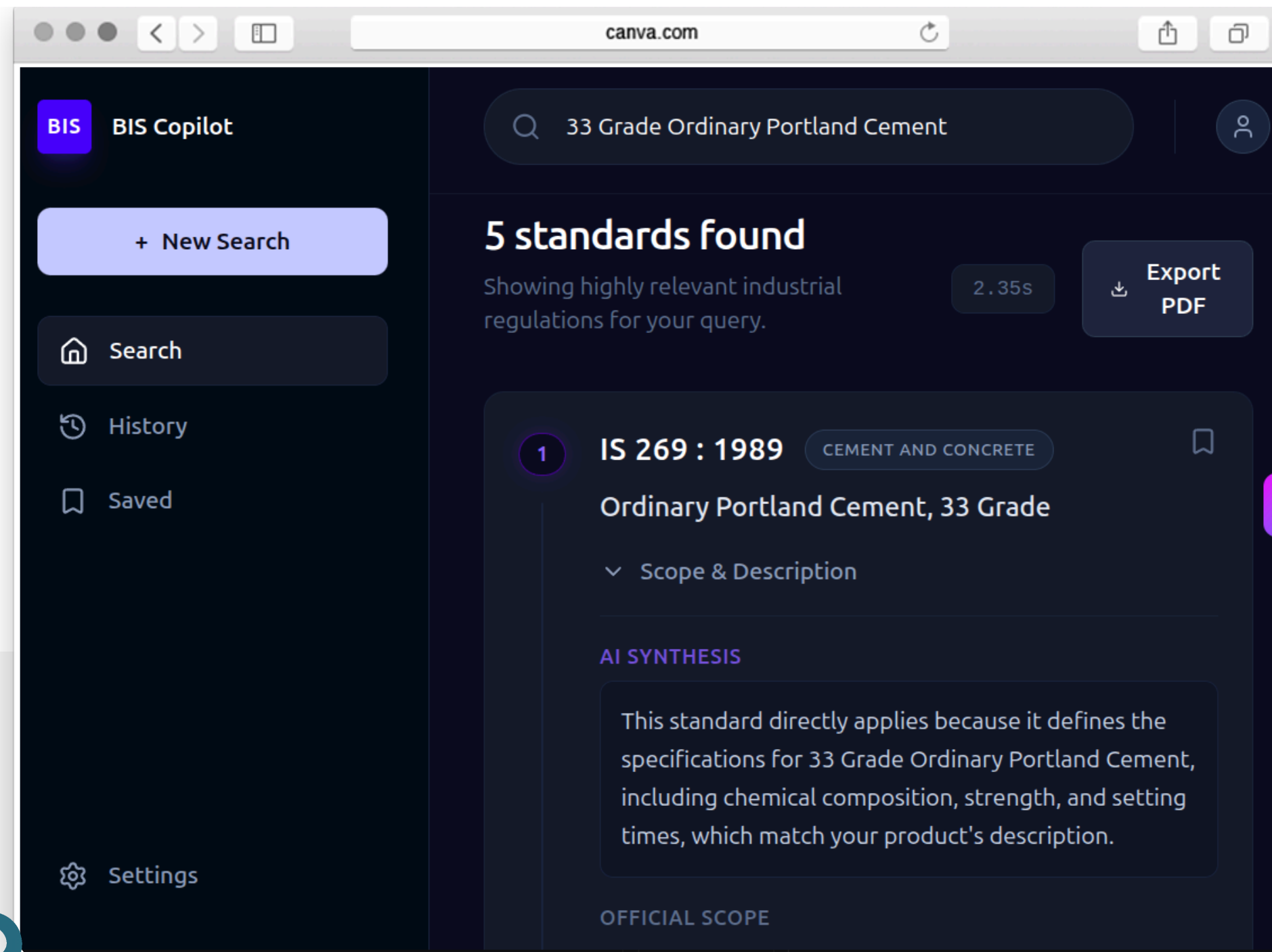
End-to-End RAG Pipeline



STRUCTURED PARSING > BLIND CHUNKING

- The Flaw of Standard RAG: Blindly chopping PDFs into 512-token chunks destroys context and leads to hallucinated metadata.
- Our Approach: We built a custom state-machine parser to handle SP 21's tricky two-column layout.
- JSON Objects, Not Chunks: Every standard is extracted as a fully typed entity (is_code, title, scope, category).
- Hybrid Retrieval:
 1. *Dense Search*: BGE-small-en-v1.5 embeddings for semantic matching.
 2. *Sparse Search*: BM25Okapi for exact IS code and keyword matching.
 3. *Fusion*: Reciprocal Rank Fusion (RRF) combines both lists for ultimate accuracy.

BIS COPILOT IN ACTION



- Fast Mode vs. AI Rationale Mode: Toggle between ~50ms pure retrieval and LLM-augmented explanations.
- Accurate Code Extraction: Handles complex sub-parts seamlessly.
- Instant PDF Generation: Converts the search session into a branded, printable compliance document.



Proven Accuracy on the Public Test Set

- HIT RATE @3: 100% (TARGET: >80%)
- MRR @5: 0.8833 (TARGET: >0.7)
- AVERAGE LATENCY: ~67MS (TARGET: <5 SECONDS)
- HALLUCINATIONS: 0%. BECAUSE IS CODES ARE STRICT METADATA AND NEVER GENERATED BY THE LLM, THE SYSTEM CANNOT HALLUCINATE A STANDARD.



EMPOWERING THE BACKBONE OF INDIA'S ECONOMY

- *Time Saved:* Reduces discovery time from weeks to seconds.
- *Cost Efficiency:* Eliminates the need for expensive third-party compliance consultants for initial standard identification.
- *Confidence:* The PDF export allows manufacturers to confidently present compliance targets to floor managers and auditors.
- *Scalable:* The structured JSON foundation allows for easy future additions like vernacular translation (Hindi, Marathi) and automated gap-analysis.



TEAM: PROJECT PHOENIX

Vedant Vankar: Backend Developer

Gitanjali Pandey: Frontend Developer

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